OS & SP

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SKILL – 08

1.Skill8.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 100

/* Variable Expansion */
char *expand_variable(const char *name)
{
    char *value = getenv(name);
```

```
    if (value == NULL)
        return strdup("");

    return strdup(value);
}

/* Expand $VARIABLE inside a token */
char *expand_token(const char *token)
{
    char result[MAX_INPUT];
    int i = 0;
    int j = 0;

    while (token[i] != '\0' && j < MAX_INPUT - 1) {
        if (token[i] == '$') {
            i++;
```

```
char variable[256];
int v = 0;

while (
    token[i] != '\0' &&
    (
        (token[i] >= 'A' && token[i] <= 'Z') ||
        (token[i] >= 'a' && token[i] <= 'z') ||
        (token[i] >= '0' && token[i] <= '9') ||
        token[i] == '_'
    )
) {
    if (v < 255)
        variable[v++] = token[i];

    i++;
}
```

```
        variable[v] = '\0';

        char *value = expand_variable(variable);

        for (int k = 0;
             value[k] != '\0' && j < MAX_INPUT - 1;
             k++) {
            result[j++] = value[k];
        }

        free(value);
    } else {
        result[j++] = token[i++];
    }
}

result[j] = '\0';
return strdup(result);
```

```
}

/* Parsing */
int parse_command(char *input, char *args[])
{
    int argc = 0;
    char *token = strtok(input, " \t");

    while (token != NULL && argc < MAX_ARGS - 1) {
        args[argc++] = token;
        token = strtok(NULL, " \t");
    }

    args[argc] = NULL;
    return argc;
}

/* Built-in help */
```

```
void builtin_help(void)
{
    printf("\nBuilt-in commands:\n");
    printf(" cd <directory>  - change directory\n");
    printf(" pwd            - show current directory\n");
    printf(" set NAME VALUE  - set environment variable\n");
    printf(" getenv NAME     - display variable\n");
    printf(" help           - show help\n");
    printf(" exit           - exit program\n\n");
}

/* Display current directory */
int builtin_pwd(void)
{
    char directory[MAX_INPUT];

    if (getcwd(directory, sizeof(directory)) == NULL) {
        perror("getcwd");
```

```
        return 1;
    }

    printf("%s\n", directory);
    return 0;
}

/* Change directory */
int builtin_cd(char *args[], int argc)
{
    if (argc < 2) {
        printf("Usage: cd <directory>\n");
        return 1;
    }

    if (chdir(args[1]) != 0) {
        perror("cd");
        return 1;
```

```
    }

    return 0;
}

/* Set environment variable */
int builtin_set(char *args[], int argc)
{
    if (argc < 3) {
        printf("Usage: set NAME VALUE\n");
        return 1;
    }

    if (setenv(args[1], args[2], 1) != 0) {
        perror("setenv");
        return 1;
    }
```

```
    printf("Variable %s set to %s\n", args[1], args[2]);
    return 0;
}

/* Display environment variable */
int builtin_getenv(char *args[], int argc)
{
    if (argc < 2) {
        printf("Usage: getenv NAME\n");
        return 1;
    }

    char *value = getenv(args[1]);

    if (value == NULL) {
        printf("%s is undefined.\n", args[1]);
    } else {
        printf("%s = %s\n", args[1], value);
```

```
    }

    return 0;
}

/* Built-in dispatcher */
int handle_builtin(char *args[], int argc)
{
    if (argc == 0)
        return 1;

    if (strcmp(args[0], "help") == 0) {
        builtin_help();
        return 1;
    }

    if (strcmp(args[0], "pwd") == 0) {
        builtin_pwd();
```

```
    return 1;
}

if (strcmp(args[0], "cd") == 0) {
    builtin_cd(args, argc);
    return 1;
}

if (strcmp(args[0], "set") == 0) {
    builtin_set(args, argc);
    return 1;
}

if (strcmp(args[0], "getenv") == 0) {
    builtin_getenv(args, argc);
    return 1;
}
```

```
    return 0;
}

/* Main */
int main(void)
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    char *expanded[MAX_ARGS];


printf("==========================================\n
");
    printf("        OSSP PROGRAM 8\n");
    printf("    Variable Expansion + Built-ins\n");

printf("==========================================\n
");
```

```
printf("\nType 'help' for commands.\n\n");

while (1) {
    printf("shell8> ");
    fflush(stdout);

    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0)
        continue;

    if (strcmp(input, "exit") == 0)
        break;

    int argc = parse_command(input, args);
```

```
if (argc == 0)
    continue;

/* Expand variables in every token */
for (int i = 0; i < argc; i++) {
    expanded[i] = expand_token(args[i]);

    if (expanded[i] == NULL) {
        perror("strdup");
        for (int j = 0; j < i; j++)
            free(expanded[j]);
        exit(EXIT_FAILURE);
    }
}

expanded[argc] = NULL;
```

```
printf("\nExpanded tokens:\n");

for (int i = 0; i < argc; i++) {
   printf(" argv[%d] = [%s]\n", i, expanded[i]);
}

/* Built-in command dispatch */
if (handle_builtin(expanded, argc)) {
   for (int i = 0; i < argc; i++)
      free(expanded[i]);

   printf("\n");
   continue;
}

/* External command */
pid_t pid = fork();
```

```
if (pid < 0) {
    perror("fork");
} else if (pid == 0) {
    execvp(expanded[0], expanded);
    perror("execvp");
    _exit(EXIT_FAILURE);
} else {
    int status;

    if (waitpid(pid, &status, 0) == -1) {
        perror("waitpid");
    } else if (WIFEXITED(status)) {
        printf("Child exited with status %d\n",
              WEXITSTATUS(status));
    }
}

for (int i = 0; i < argc; i++)
```

```
        free(expanded[i]);


    printf("\n");
  }


  printf("Program 8 terminated.\n");
  return 0;
}
```

OUTPUT :

